

Unit: DAM
2025-2026

DAM Project *"Campus Connect - A Student Utility App"*

Campus Connect is a Flutter-based mobile app that streamlines student life on campus. It offers tools like class schedules, exam notifications, a digital ID card, event updates, and peer-to-peer chat — all in one app.

Actors (SAME APPLICATION, same login page, use one admin and at least two teachers with predefined accounts, use few predefined courses as well)

Admins - Admin(s) manage the students, they can accept new registrations, affect groups to teachers/students, can delete students and upload files/schedules/timetables (for students to check, can be simple images), and can assign courses to teachers and student groups.

Teachers - Teachers can mark each student of their group; can also upload files for their course, and mark attendance per week.

Students - Each new student have to register in order to use the application, the registration has to be validated by an admin. Once signed in, he should be able to check his profile (in form of a student card, some information can be changed). Each student can check his group schedule, his marks per course, and

Some Hints:

- Flutter framework must be used.
- You can use SQL database with flutter directly.
- Note that you can rely on your own imagination, you do not have to follow these sprints.
- You may use **any type of database** you want (firebase, NoSQL, SQL...etc. **except for SQLite**),
As a reminder, **REST API SQL** Connection and usage will be presented as a course chapter, and will be **better marked**.
- Note that Sprints 1, 2 are mandatory to get the minimum mark. Doing sprints from 3 to 4 will increase the obtained mark obviously.

The project will be marked as 11/20 of the total TP mark. For more transparency, each sprint has its own respective mark.

Sprint 1: Front end and UI (3 points)

- Design a basic, user-friendly interface for each actor.
- Design a student card for each student with his information.
- Design a login page and a home for each actor.

Sprint 2: Functions (2.5 points)

- Design a full functional admin panel to manage students (registrations, affect groups and their courses, upload timetable/schedules) and to manage teachers (affect group, affect course).
- Design pages for student to check their groups, their marks, their timetables, their courses and all different files that might concern them.
- Design pages for the teacher to check his courses (upload and check files as well) and modify marks for them.

Sprint 3: Backend (2.5 points)

- Register users in the database and develop a login/signup process using the database.
- Save and fetch all information from the database
- Modify data from the database when using the app and upload images on server.
- Note that you do not have to fully do all tables; you may do two tables for instance and get marked partially for them.

Sprint 4: Enhanced Search and Responsive design (1 point)

- Develop a search bar with predictive text and suggestions, so that students/admin may find courses and information quickly.
- Make your design and UI responsive!

Two (2) Marks are to be awarded for the UI design, creativity, and other non-mentioned used APIs and functions (examples: notifications, calendars, email sending and verification...etc.)

The project is to be returned after vacation and before exams, exact date will be sent later but probably on the first week. Feel free to email me your concerns if you have any.

Each project should contain no more than Three (3) students; all **students should be present during the project's consultation.**

Students can be from different groups.